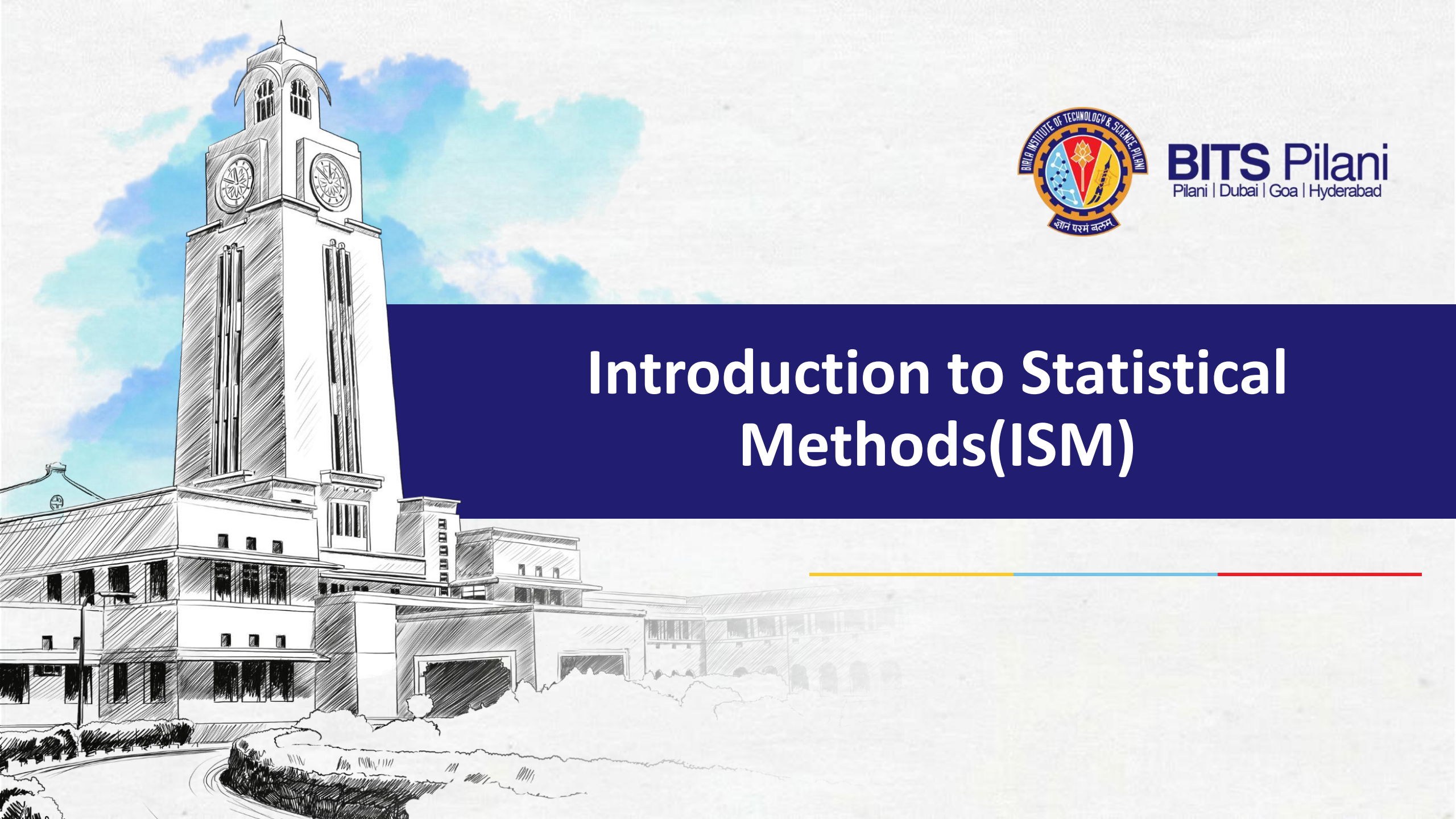
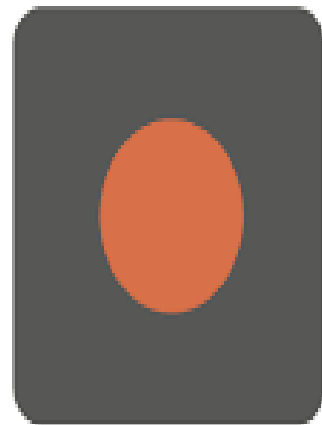




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Introduction to Statistical Methods(ISM)

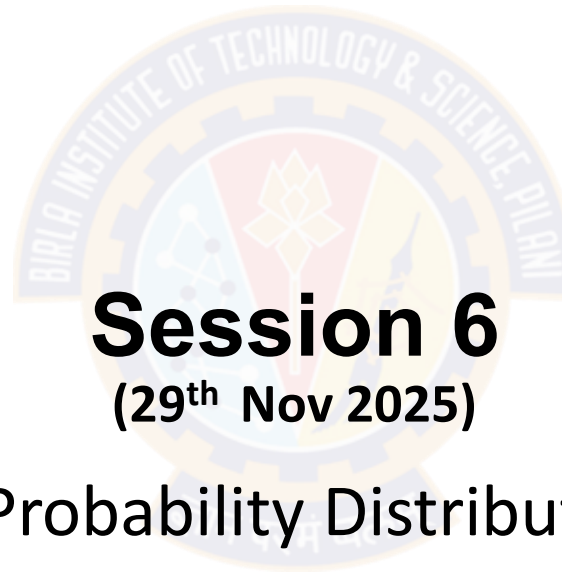
IMP Note to Self



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Session 6

(29th Nov 2025)

Probability Distributions

Revise

- Descriptive Statistics
- Review of Set Theory, Counting Principles
- Probability, Axioms of Probability
- Addition and Multiplication Rule of Probability
- Conditional Probability and Independent events
- Total Probability, Bayes Theorem and Naïve Bayes theorem
- **Random Variable, Probability Distribution**
- **Joint Probability Distribution**

Agenda

- **Important Probability Distributions:-**
 - ❖ Bernoulli
 - ❖ Binomial
 - ❖ Poisson
 - ❖ Normal distributions.
 - ❖ Introduction to t-distribution
 - ❖ Introduction to F-Distribution and
 - ❖ Introduction to Chi Square distributions



Bernoulli Distribution

- A random experiment that can only have an outcome of either 1 or 0 is known as a Bernoulli trial.
- In other words, the random variable can be 1 with a probability p or it can be 0 with a probability $(1 - p)$. Such an experiment is called a **Bernoulli trial**.
- A discrete probability distribution wherein the random variable can only have 2 possible outcomes is known as a **Bernoulli Distribution**.

Bernoulli Distribution

- Bernoulli Distribution can be used to describe events that can only have two outcomes, that is, success or failure.
- Bernoulli distribution is a special case of binomial distribution when only 1 trial is conducted.

•Example:

1) A pass or fail exam can be modeled by a Bernoulli Distribution.

2) A basketball player can shoot a ball into the basket with a probability of 0.6. What is the probability that he misses the shot?

Solution: We know that success probability $P(X = 1) = p = 0.6$

Thus, probability of failure is $P(X = 0) = 1 - p = 1 - 0.6 = 0.4$

The probability he misses the shot is 0.4.

Bernoulli Distribution

Definition

A random variable 'X' is said to have Bernoulli distribution if its probability mass function is given by

$$p(x) = \begin{cases} p^x q^{1-x}, & x = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$$

This represents a single trial with two possible outcomes (usually labeled as success/failure or 1/0) where success has a probability p and failure has a probability $q = 1 - p$.

Mean & Variance of Bernoulli distribution

The expected value of a Bernoulli random variable X is $E[X] = p$

Proof:

$$\begin{aligned} E[X] &= \sum_{x=0,1} xp(x) \\ &= 0 \cdot p(0) + 1 \cdot p(1) \\ &= 0 + 1 \cdot (p) \\ &= p \end{aligned}$$

The variance of a Bernoulli random variable X is $\text{Var}[X] = pq$

Proof:

$$\begin{aligned} E[X^2] &= \sum_{x=0,1} x^2 p(x) \\ &= 0^2 \cdot p(0) + 1^2 \cdot p(1) \\ &= 0 + 1 \cdot p \\ &= p \end{aligned}$$

$$\text{Var}[X] = E[X^2] - E[X]^2 = p - p^2 = p(1 - p) = pq$$

Bernoulli distribution: Applications

1. **Quality Control:** In manufacturing, every product undergoes quality checks. Bernoulli Distribution helps assess whether a product passes (success) or fails (failure) the quality standards.
2. **Market Research:** Bernoulli Distribution is useful in surveys and market research when dealing with yes/no questions.
3. **Risk Assessment:** In the context of risk management, the Bernoulli Distribution can be applied to model events with binary outcomes, such as a financial investment succeeding (success) or failing (failure).
4. **Marketing Campaigns:** Businesses use Bernoulli Distribution to measure the effectiveness of marketing campaigns.

Binomial Distribution

Binomial distribution is a discrete probability distribution.

Binomial distribution will be applied under the following experimental conditions

- 1) The number of trials (n) is finite
- 2) The trials are independent of each other
- 3) The probability of success p is constant for each trial.
- 4) Each trial results in two mutually exclusive events known as success and failure.

Binomial Distribution

Definition

A random variable 'X' is said to have Binomial distribution if its probability mass function is given by

$$p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, & x = 0, 1, 2, 3, \dots, n \\ 0, & \text{elsewhere} \end{cases}$$

It describes the number of successes in a fixed number of independent Bernoulli trials. It involves parameters n (number of trials) and p (probability of success).

Binomial Distribution

Binomial Distribution

n = # of trials

1-p = probability of failure

$${}^n C_x p^x (1-p)^{n-x}$$

x = # of success

p = probability of success

Mean & Variance

The expected value of a Binomial random variable X is $E[X] = np$

Proof:

$$\begin{aligned} E(x) &= E\left[\sum_{i=1}^n Y_i\right] \text{ (representation as a sum of } n \text{ independent Bernoulli r.v.)} \\ &= \sum_{i=1}^n E[Y_i] \text{ (linearity of the expected value)} \\ &= \sum_{i=1}^n p \text{ (expected value of a Bernoulli r.v.)} \\ &= np \end{aligned}$$

The variance of a Binomial random variable X is $\text{Var}[X] = npq$

Proof:

$$\begin{aligned} \text{Var}(x) &= \text{Var}\left[\sum_{i=1}^n Y_i\right] \text{ (representation as a sum of } n \text{ independent Bernoulli r.v.)} \\ &= \sum_{i=1}^n \text{Var}[Y_i] \text{ (linearity of Variance of independent variable)} \\ &= \sum_{i=1}^n pq \text{ (variance of a Bernoulli r.v.)} \\ &= npq \end{aligned}$$

Binomial Distribution-Check:

Which conditions for the binomial distribution, if any, fail to hold in each of the following situations?

(a) The number of persons getting Corona in a room of 30 persons.

Ans: Getting corona is not independent. So, it is not binomial.

(b) Consider 2 out of 20 projectors are defective. We randomly select 3 for testing. Is this a binomial experiment?

Ans: If two defective pieces are in the first two selection, then the result of third is known already. Thus, the event is not independent. So, this is not binomial.

Example 1:

In a particular restaurant, an average of 3 out of every 5 customers ask for water with their meal. A random sample of 10 customers is selected. Find the probability that

- (i) Exactly 6 ask for water with their meal,
- (ii) Less than 9 ask for water with their meal.
- (iii) No one asks for water with their meal?
- (iv) At most 2 ask for water with their meal?
- (v) At least 3 ask for water with their meal?

Solution:

$$n = 10, p = 3/5 = 0.6, q = 1 - p = 0.4$$

$$(i) p(6) = 0.251$$

$$(ii) p(\text{less than } 9) = 1 - [p(9) + p(10)] = 0.954$$

$$(iii) p(0) = 0.0001$$

$$(iv) p(\text{at most } 2) = p(0) + p(1) + p(2) = 0.0001 + 0.0016 + 0.0106 \\ = 0.0123$$

$$(v) p(\text{at least } 3) = p(3) + \dots + p(10) \\ = 1 - [p(0) + p(1) + p(2)] = 0.9877$$

Understandings

Phrase	<i>Math Symbol</i>
“at least”	$\geq$
“more than” or “greater than”	$>$
“fewer than” or “less than”	$<$
“no more than”	$\leq$
“exactly”	$=$

Example 2:

Ten coins are thrown simultaneously. what is the probability of getting 3 or fewer heads?

Solution:

$$P(x \leq 3) = \sum p(x) = P(0) + P(1) + P(2) + P(3)$$

$$= {}^{10}C_0(0.5)^0(0.5)^{10} + {}^{10}C_1(0.5)^1(0.5)^9 + {}^{10}C_2(0.5)^2(0.5)^8 + {}^{10}C_3(0.5)^3(0.5)^7$$

$$= (0.5)^{10} [{}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3]$$
$$= (0.5)^{10} [1 + 10 + 45 + 120] = 176/1024 = 0.1718$$

$$P(x \leq 3) = \sum p(x) = 0.1718$$

Example 3:

A pair of dice is rolled 5 times. If getting the product of 6 is considered as a success. Find the probability of getting at least 4 successes.

Solution:

The formula for the above probability is given by:

$$P(X = x) = {}^nC_x p^x q^{n-x}$$

Here, $n = 5$

p is the probability of getting product 6

$$p = 4/36 = 1/9$$

$$q = 1 - p = 8/9$$

$$\Rightarrow P(X \geq 4) = P(X = 4) + P(X = 5)$$

$$\Rightarrow P(X \geq 4) = {}^5C_4 (1/9)^4 (8/9)^{5-4} + {}^5C_5 (1/9)^5 (8/9)^{5-5}$$

$$\Rightarrow P(X \geq 4) = 5(1/9)^4 (8/9) + (1/9)^5$$
$$= 0.000694335$$

Example 4:

Each of five questions on a multiple-choice examination has four choices, only one of which is correct. The student is attempting to guess the answers. The random variable X is the number of questions answered correctly. What is the probability that the student will get

- a) exactly three correct answers?
- b) at most three correct answers?
- c) at least one correct answer?
- d) Find Mean and variance.



Solution:

$$p=1/4, n=5, q=3/4;$$

$$a) P(X=3) = 45/512 = 0.08789$$

$$b) P(X \leq 3) = 63/64;$$

$$c) P(X \geq 1) = 0.7627$$

$$d) \text{Mean} = 5/4$$
$$\text{Variance} = 15/16.$$



Example 5:

➤ Find the binomial distribution if the mean is 6 and variance is 4.

• **Solution:** $n=18$, $p=1/3$;



Example 6:

In a large number of parts manufactured by a machine , the mean number of defectives in a sample of 20 is 2. Out of 1000 such samples , how many would be expected to contain atleast 3 defective parts.

Sol: Given $n=20$; mean $= \mu = 2$, but $\mu = np \Rightarrow 2 = 20p \therefore p = 0.1$

$$\therefore P(x) = {}^{20}C_x (0.1)^x (0.9)^{20-x}$$

Therefore the probability that there are at least 3 defective parts in a sample of 20 is

$$P(X \geq 3) = 1 - P(X < 3) = 1 - [P(0) + P(1) + P(2)]$$

$$\therefore P(X \geq 3) = 0.323$$

Hence out of 1000 samples we get $0.323 \times 1000 = 323$ samples

Recall : Binomial Distribution

A **binomial experiment** is a probability experiment that satisfies the following conditions.

1. The experiment is **repeated** for a fixed number of trials, where each trial is **independent** of other trials.
2. There are only **two possible outcomes** of interest for each trial. The outcomes can be classified as a success (S) or as a failure (F).
3. The probability of a success $P(S)$ is the same for each trial.
4. The random variable x counts the number of successful trials.

Poisson Distribution

➤ Poisson distribution is suitable for **rare events** for which the probability of occurrence 'p' is very small and the number of trials 'n' is very large.

➤ Also, binomial distribution can be approximated by Poisson distribution when $n \rightarrow \infty$ and $p \rightarrow 0$ such that $\lambda = np = \text{constant}$

The approximation is more accurate if $n \geq 100$ and as long as $np \leq 10$

Poisson Distribution

Definition

A random variable 'X' is said to have Poisson distribution if its probability mass function is given by

$$p(x) = \begin{cases} e^{-\lambda} \frac{\lambda^x}{x!}, & x = 0, 1, 2, \dots \\ 0, & \text{elsewhere} \end{cases}$$

- Poisson distribution is the discrete probability distribution of a discrete random variable X, which may not have upper bound. It is defined for non negative values of x .

Example 6:

•Telephone calls enter a college switchboard on the average of two every three minutes. What is the probability of 5 or more calls arriving in a 9-minute period?

•Solution:

$$\lambda = 2 \cdot 9/3 = 6, \quad P(X \geq 5) = 1 - P(X \leq 4) \doteq 0.715$$

Example 7:

If 3% of electronic units manufactured by a company are defective. Find the probability that in a sample of 200 units, less than 2 bulbs are defective.

Solution:

The probability of defective units $p = 3/100 = 0.03$

Give $n = 200$.

We observe that p is small and n is large here. Thus, it is a Poisson distribution.

Mean $\lambda = np = 200 \times 0.03 = 6$

$P(X = x)$ is given by the Poisson Distribution Formula as $(e^{-\lambda} \lambda^x) / x!$

$$\begin{aligned} P(X < 2) &= P(X = 0) + P(X = 1) \\ &= (e^{-6} 6^0) / 0! + (e^{-6} 6^1) / 1! \\ &= e^{-6} + e^{-6} \times 6 \\ &= 0.00247 + 0.0148 \end{aligned}$$

$$P(X < 2) = 0.01727$$

Example 8:

The number of customers arriving at a cafeteria at an average rate of 0.3 per minute.

- (a) Find the probability that exactly 2 customers arrive in a 10-minute span.
- (b) Find the probability that 2 or more customers arrive in a 10-minute span.
- (c) Find the probability that exactly one customer arrives in a 5-minute span and one customer arrives in the next 5-minute span.

Solution:

Solution

We use a Poisson process with a parameter λt , $\lambda t = 0.3 \times 10 = 3$ in items a) and b) .

a)

$$P(2 \text{ customers in 10 minute span}) = \frac{e^{-0.3 \cdot 10} (0.3 \cdot 10)^2}{2!} = 0.224.$$

b)

$$\begin{aligned} P(2 \text{ or more in 10 minute span}) &= 1 - P(0) - P(1) \\ &= 1 - \frac{e^{-0.3 \cdot 10} (0.3 \cdot 10)^0}{0!} - \frac{e^{-0.3 \cdot 10} (0.3 \cdot 10)^1}{1!} = 0.8. \end{aligned}$$

c)

$$\begin{aligned} P(\text{one customer in 5 minute and one customer in the next minute}) \\ = \frac{e^{-0.3 \cdot 5} (0.3 \cdot 5)^1}{1!} \cdot \frac{e^{-0.3 \cdot 1} (0.3 \cdot 1)^1}{1!} = 0.074. \end{aligned}$$

Mean and Variance

Mean:

$$\mu = \lambda$$

For a Poisson random variable, the variance and mean are the same!

Variance and Standard Deviation

$$\sigma^2 = \lambda$$

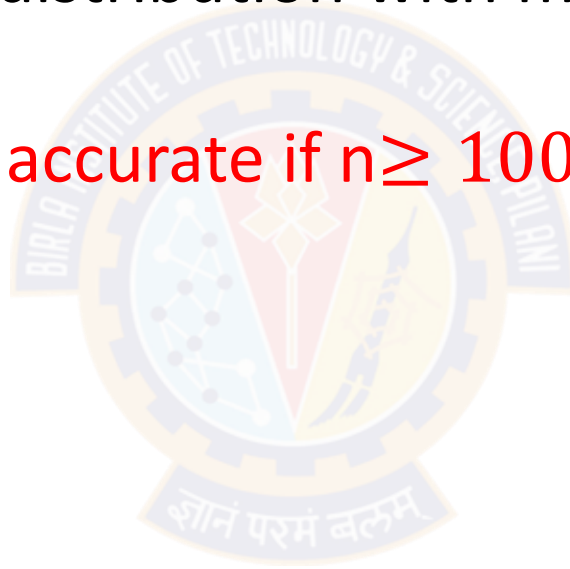
$$\sigma = \sqrt{\lambda}$$

where λ = expected number of occurrences in a given experiment.

Poisson approximation to Binomial distribution :

When n is large and p is small, Binomial probabilities can be approximated by Poisson's distribution with mean $\lambda=np$.

The approximation is more accurate if $n \geq 100$
and as long as $np \leq 10$



Example 9:

It is known that 5% of the books bound at a certain bindery have defective bindings. Find the probability that 2 of 100 books bound by this bindery will have defective bindings using

- (a) The formula for the binomial distribution;
- (b) The Poisson approximation to the binomial distribution

Solution:

(a) Here $x = 2$, $n = 100$, and $p = 0.05$. using the formula for the binomial distribution, we get

$$b(2;100,0.05) = \binom{100}{2} (0.05)^2 (0.95)^{98} = 0.081$$

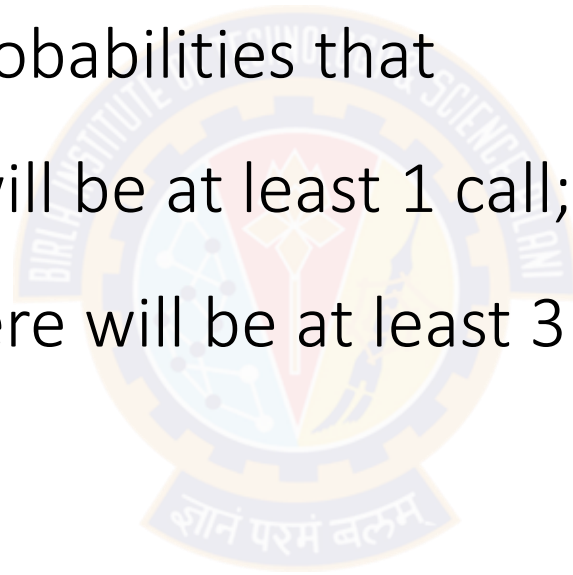
(b) Substituting $x = 2$ and $\lambda = np = 100(0.05) = 5$ into the formula for the Poisson distribution, we get

$$f(2;5) = \frac{5^2 \cdot e^{-5}}{2!} = 0.084$$

Example 10:

Given that a switch board of a consultant's office receives on the average 0.6 calls per minute, find the probabilities that

- (a) in a given minute there will be at least 1 call;
- (b) in a 4-minute interval there will be at least 3 calls.



Solution:

- a) Here,

$$\lambda = 0.6 \text{ and } t = 1 \therefore p(x, \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

$$p(x, \lambda t) = \frac{e^{-0.6} (0.6)^x}{x!}$$

$$P(X \geq 1) = 1 - P(X < 1) = 1 - P(0)$$

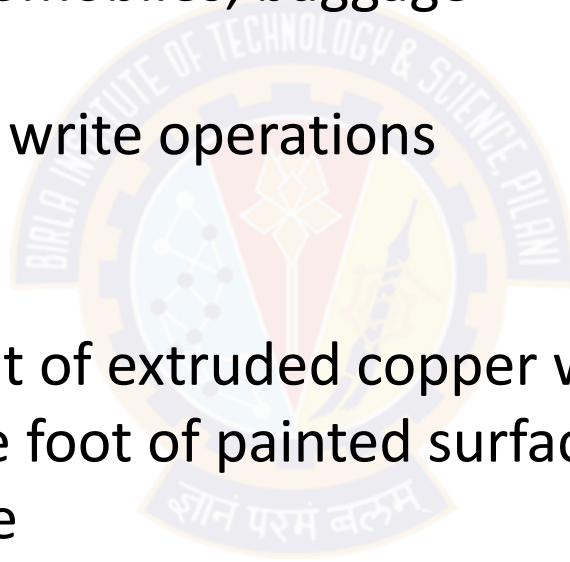
- b) $= 1 - e^{-0.6} = 0.4512$

$$\lambda = 0.6 \text{ and } t = 4 \therefore p(x, \lambda t) = \frac{e^{-2.4} (2.4)^x}{x!}$$

$$P(X \geq 3) = 1 - P(X < 3) = 1 - [P(0) + P(1) + P(2)]$$
$$= 0.4303$$

Poisson Distribution: Applications

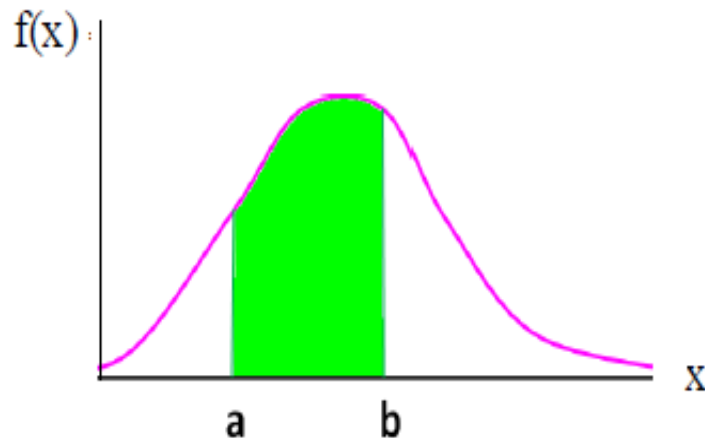
- Arrivals at queuing systems:
 - ✓ Airports : people, airplanes, automobiles, baggage
 - ✓ Banks: people, loan applications
 - ✓ computer file servers : read and write operations
- Defects in manufactured goods:
 - ✓ number of defects per 1,000 feet of extruded copper wire
 - ✓ number of blemishes per square foot of painted surface
 - ✓ number of errors per typed page



RECALL: Continuous Random Variables

A continuous random variable can assume any value in an interval on the real line or in a collection of intervals.

The probability of the random variable assuming a value within some given interval from x_1 to x_2 is defined to be the area under the graph of the **probability density function** between x_1 and x_2



Example:

- Height of students in a class
- Amount of ice tea in a glass
- Change in temperature throughout a day
- Price of a car in next year



Continuous Probability Distributions:

Normal Distribution:

Normal distribution is a general distribution. Any information such as heights, weights, lengths, widths, sizes, profits, prices, wages, time etc. containing sufficiently large data with mean and variance follows normal distribution.

The distribution of errors of repeated measurements follows normal distribution.

Normal Distribution

A continuous random variable X which assumes all possible values in the entire real space, i.e., $-\infty < x < \infty$ is said to follow normal distribution with two parameters such as mean μ and variance σ^2 if it has the probability density function given by

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \begin{matrix} -\infty < x < \infty \\ -\infty < \mu < \infty, \sigma > 0 \end{matrix}$$

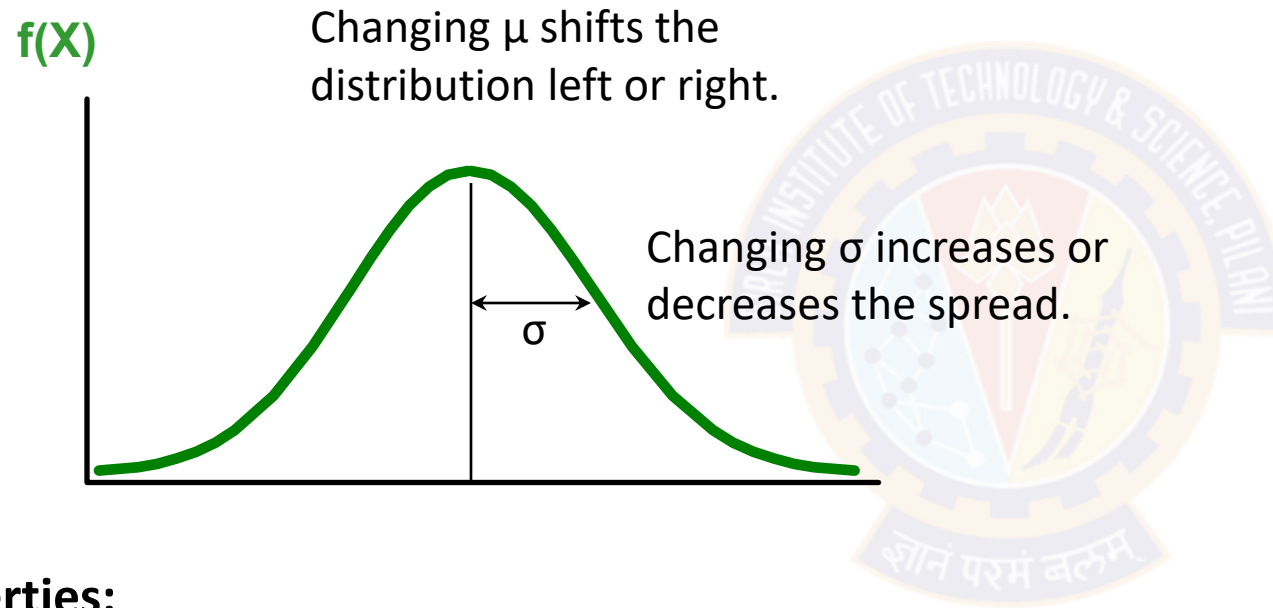
μ = mean

σ = standard deviation

π = 3.14159

e = 2.71828

Normal Distribution

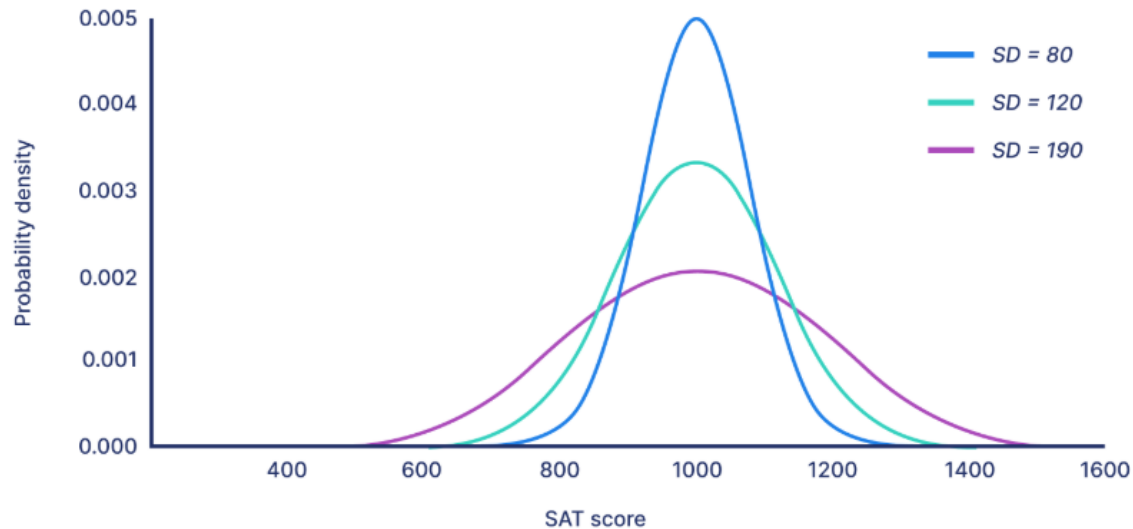


Properties:

1. Normal curve is bell shaped and symmetric about the mean
2. Mean = Mode = Median
3. Total area under normal curve is equal to 1
4. Normal curve approaches but never touches the x axis as it extends farther and farther away from the mean

Normal Distribution

Normal distributions with different standard deviations

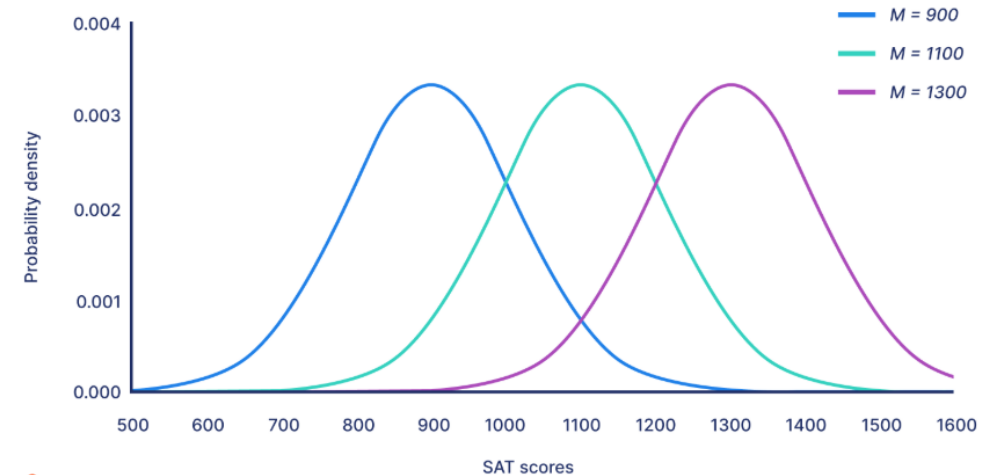


Empirical rule

The **empirical rule**, or the 68-95-99.7 rule, tells you where most of your values lie in a normal distribution:

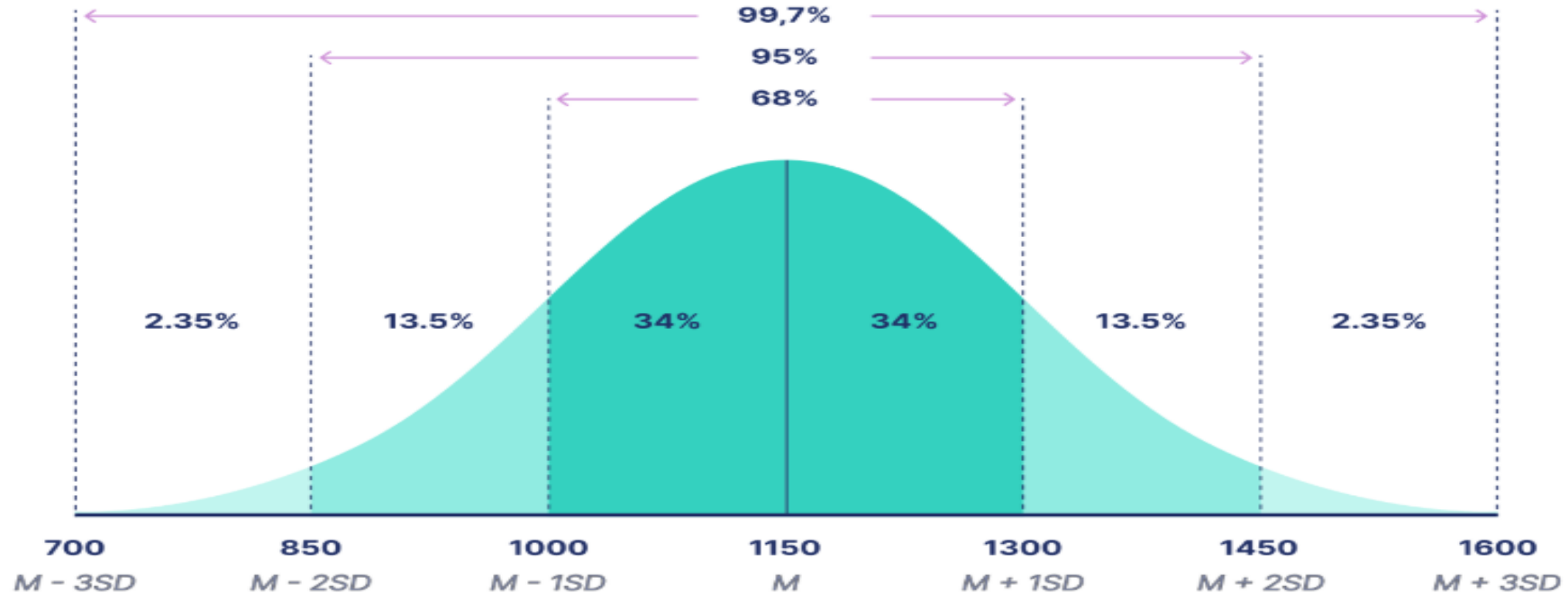
- Around 68% of values are within 1 standard deviation from the mean.
- Around 95% of values are within 2 standard deviations from the mean.
- Around 99.7% of values are within 3 standard deviations from the mean.

Normal distributions with different means



Area under normal curve

Using the empirical rule in a normal distribution



No matter what μ and σ are, the area between $\mu - \sigma$ and $\mu + \sigma$ is about 68%; the area between $\mu - 2\sigma$ and $\mu + 2\sigma$ is about 95%; and the area between $\mu - 3\sigma$ and $\mu + 3\sigma$ is about 99.7%. Almost all values fall within 3 standard deviations.

Normal Distribution: Mean and Variance

$$E(X)=\mu = \int_{-\infty}^{+\infty} x \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\text{Var}(X)=\sigma^2 = \left(\int_{-\infty}^{+\infty} x^2 \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \right) - \mu^2$$

Standard Deviation(X)= σ

Why Standard Normal Distribution?

	x	$X - \mu_x$	$(X - \mu_x)^2$	$Z = \frac{X - \mu_x}{\sigma_x}$	$(Z - \mu_z)^2$
	1	-2	4	-1.42	2.01
	2	-1	1	-0.71	0.50
	3	0	0	0.00	0.00
	4	1	1	0.71	0.50
	5	2	4	1.42	2.01
Sum	15	0	10	0	5.03
Mean	$\mu_x = 3$			$\mu_z = 0$	
SD	$\sigma_x = 1.41$			$\sigma_z = 1.00$	

$X \sim N(\mu, \sigma^2) \rightarrow Z \sim N(0, 1)$. This is true for any data size

Why Standard Normal Distribution?

- $Z = \frac{X - \mu}{\sigma}$

$$\text{Mean of } z = E(z) = E\left(\frac{x - \mu}{\sigma}\right) = \frac{1}{\sigma} [E(x) - E(\mu)] = \frac{1}{\sigma} [\mu - \mu] = 0$$

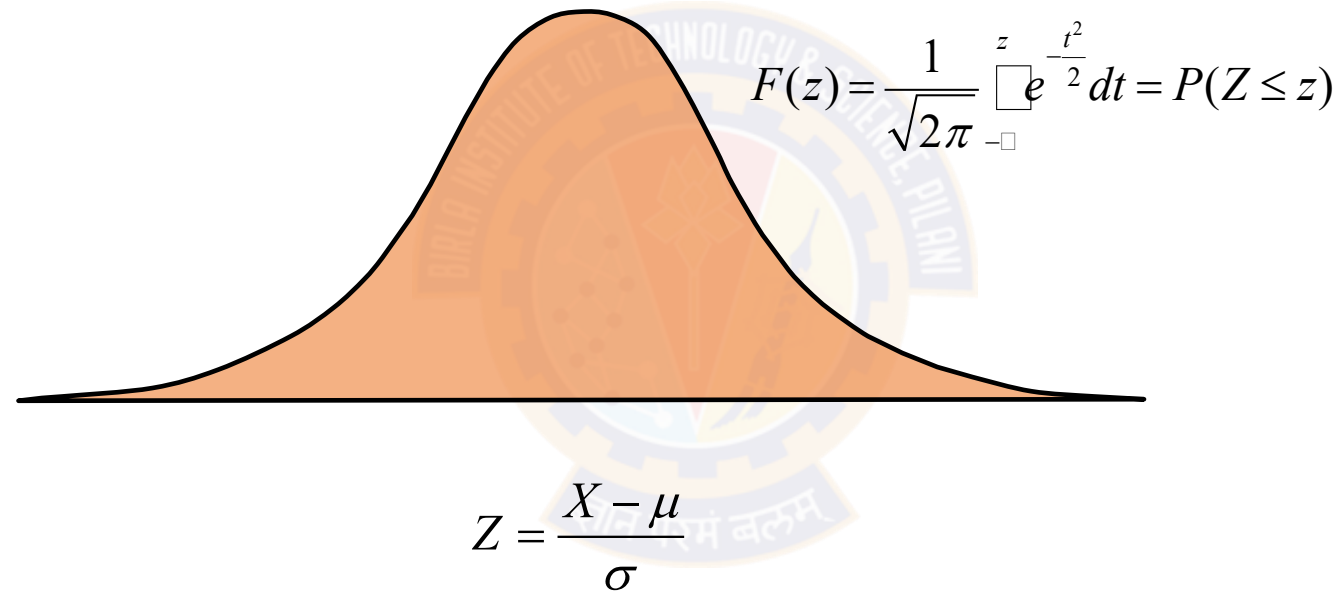
$$\begin{aligned} V(z) &= E[z - E(z)]^2 = E(z^2) \quad \text{As } E(z)=0 \\ &= E\left(\frac{x - \mu}{\sigma}\right)^2 \\ &= \frac{1}{\sigma^2} E(x - \mu)^2 \quad \text{AS } v(ax) = a^2 v(x) \\ &= \frac{1}{\sigma^2} \sigma^2 = 1 \end{aligned}$$

$X \sim N(\mu, \sigma^2) \rightarrow Z \sim N(0, 1)$. This is true for any data size.

In general, $X \sim N(\mu, \sigma^2)$ then $Y = aX + b \sim N(a\mu + b, a^2\sigma^2)$ as a linear transformation of a normally distributed variable is *still normal*

Standard Normal Distribution

The **standard normal distribution** is a normal distribution with a mean of 0 and a standard deviation of 1.



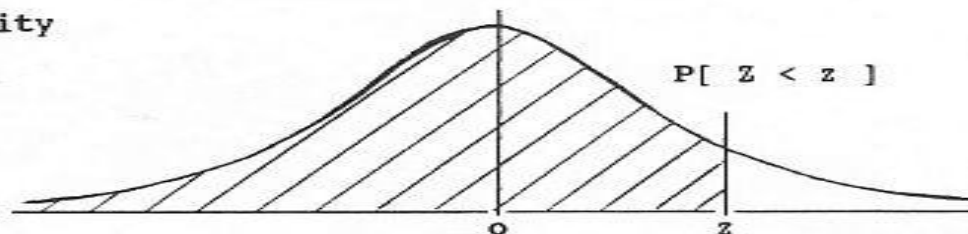
All normal distributions can be converted into the standard normal curve by subtracting the mean and dividing by the standard deviation.

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability up to the standardised normal value z i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2) dz$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9874	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
z	3.00	3.10	3.20	3.30	3.40	3.50	3.60	3.70	3.80	3.90
P	0.9986	0.9990	0.9993	0.9995	0.9997	0.9998	0.9998	0.9999	0.9999	1.0000

Example 11:

If a random variable has the standard normal distribution, find the probability that it will take on a value

a)less than 1.75 = $P(Z < 1.75) = F(1.75) = 0.9599$

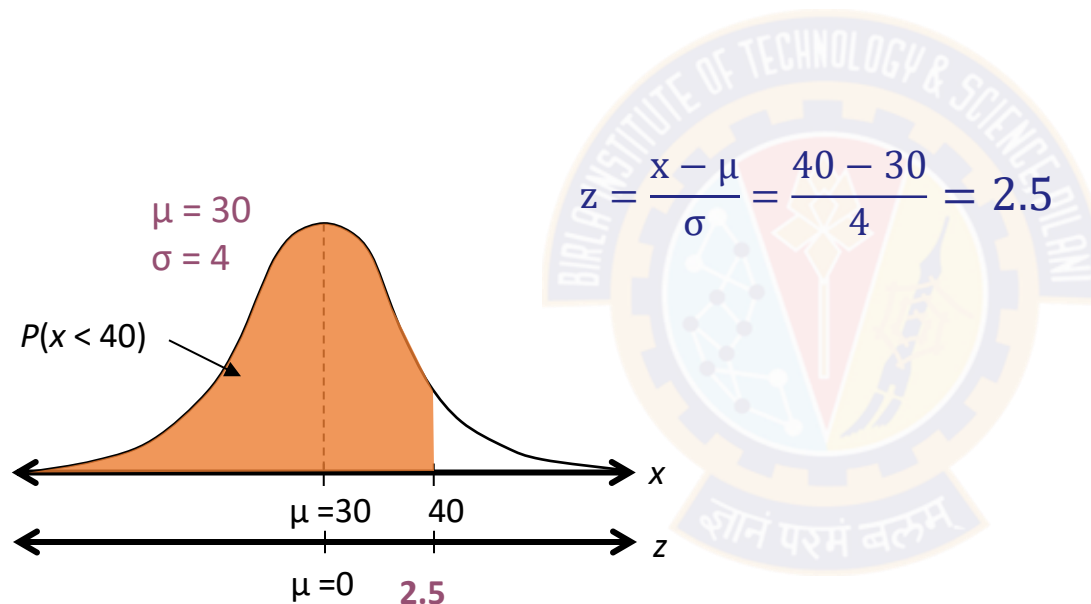
b)less than -1.25 = $P(Z < -1.25) = F(-1.25) = 1 - F(1.25) = 0.1056$

c)greater than 2.06 = $P(Z > 2.06) = 1 - F(2.06) = 1 - 0.9803 = 0.0197$

d)greater than -1.82 = $P(Z > -1.82) = 1 - F(-1.82) = 1 - 0.0344 = 0.9656$

Example 12:

Scores on a particular test are normally distributed with a mean of 30 and standard deviation of 4. What is the probability of anyone scoring less than 40?

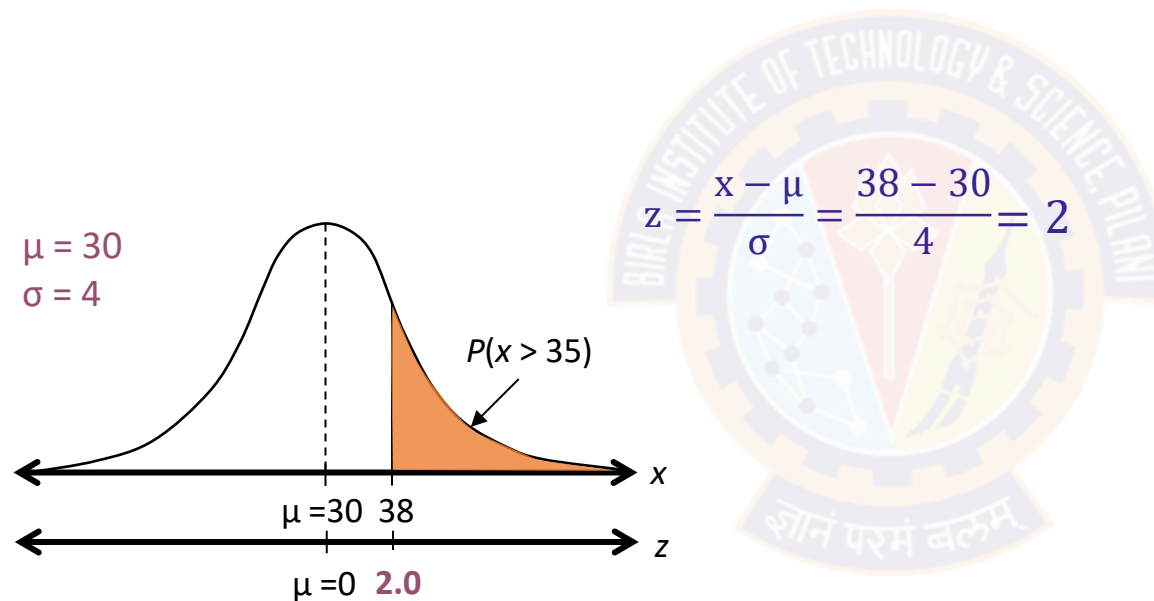


$$P(x < 40) = P(z < 2.5) = 0.9938$$

The probability that a student receives a test score less than 40 is **0.9938**.

Example 13:

Scores on a particular test are normally distributed with a mean of 30 and standard deviation of 4. What is the probability of anyone scoring greater than 38?

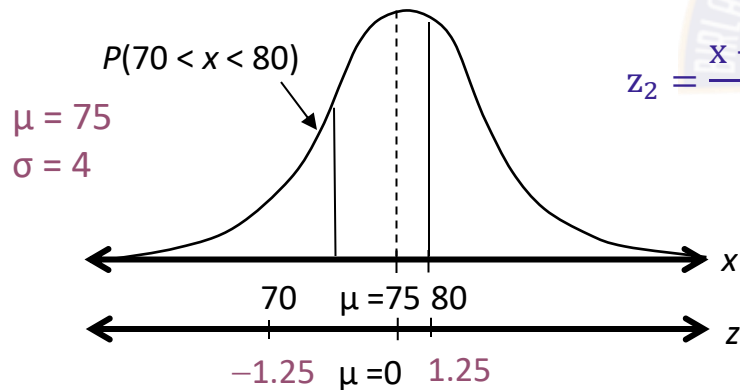


$$P(x > 38) = P(z > 2) = 1 - P(z < 2) = 1 - 0.9773 = 0.0227$$

The probability that a student receives a test score greater than 38 is 0.0227

Example 14:

The average on a statistics test was 75 with a standard deviation of 4. If the test scores are normally distributed, find the probability that a student receives a test score between 70 and 80.



$$z_1 = \frac{x - \mu}{\sigma} = \frac{70 - 75}{4} = -1.25$$

$$z_2 = \frac{x - \mu}{\sigma} = \frac{80 - 75}{4} = 1.25$$

The probability that a student receives a test score between 70 and 80 is 0.7888

$$\begin{aligned} P(70 < x < 80) &= P(-1.25 < z < 1.25) = F(1.25) - F(-1.25) \\ &= 0.8944 - 0.1056 = 0.7888 \end{aligned}$$

Example 15:

If X is a normal variate with mean 30 and SD 5. Find the probabilities that

(i) $26 \leq X \leq 40$ (ii) $X > 45$

Solution :

Here, mean $\mu = 30$ and standard deviation $\sigma = 5$

(i) When $X = 26$

$$Z = (X - \mu) / \sigma = (26 - 30) / 5 = -0.8 \text{ and}$$

$$\text{when } X = 40, Z = (40 - 30) / 5 = 2$$

$$\begin{aligned} \text{Therefore, } P(26 \leq X \leq 40) &= F(2) - F(-0.8) = F(2) - (1 - F(0.8)) \\ &= 0.9773 - 1 + 0.7881 = 0.7654 \end{aligned}$$

$$(ii) P(X \geq 45) = P(Z \geq 3) = 1 - F(3) = 1 - 0.9986 = 0.0014$$

Example 16:

Butterfly-style valves used in heating and ventilating industries have a high flow coefficient. Flow coefficient can be modeled by a normal distribution with mean $496 C_v$ and standard deviation $25C_v$. Find the probability that a valve will have a flow coefficient of at least $450C_v$

$$P(X > 450) = P[(450 - 496) / 25] = P(Z > -1.84) = 1 - F(-1.84) = 1 - 0.0329 = .9671$$

b) between 445.5 and $522C_v$

$$\begin{aligned} P(445.5 < X < 522) &= P(-2.02 < Z < 1.04) = F(1.04) - F(-2.02) = 0.8508 - 0.0217 \\ &= 0.8291 \end{aligned}$$

Normal Approximation to a Binomial Distribution

The normal distribution is used to approximate the binomial distribution **when it would be impractical to use the binomial distribution** to find a probability.

Normal Approximation to a Binomial Distribution

If $np \geq 15$ and $nq \geq 15$, then the binomial random variable x is approximately normally distributed with mean

$$\mu = np$$

and standard deviation

$$\sigma = \sqrt{npq}.$$

Normal Approximation to a Binomial Distribution

Correction for Continuity: Since the binomial distribution is discrete and the normal distribution is continuous, a continuity correction is applied. This involves adjusting the discrete binomial probability by 0.5 in either direction to account for the continuous nature of the normal distribution.

- When approximating $P(X \leq k)$, use $P(X \leq k+0.5)$
- When approximating $P(X < k)$, use $P(X < k-0.5)$
- When approximating $P(X \geq k)$, use $P(X \geq k-0.5)$
- When approximating $P(X > k)$, use $P(X > k+0.5)$.

Discrete	Continuous
$x = 6$	$5.5 < x < 6.5$
$x > 6$	$x > 6.5$
$x \leq 6$	$x < 6.5$
$x < 6$	$x < 5.5$
$x \geq 6$	$x > 5.5$

Example 17:

Decide whether the normal distribution to approximate x may be used in the following examples:

1. In a survey of 8 to 18-year-old heavy media users in the United States, 47% said they get fair or poor grades (C's or below). You randomly select forty-five 8 to 18-year-old heavy media users in the United States and ask them whether they get fair or poor grades.

Solution:

$$np = 45(0.47) = 21.15$$

$$nq = 45(0.53) = 23.85.$$

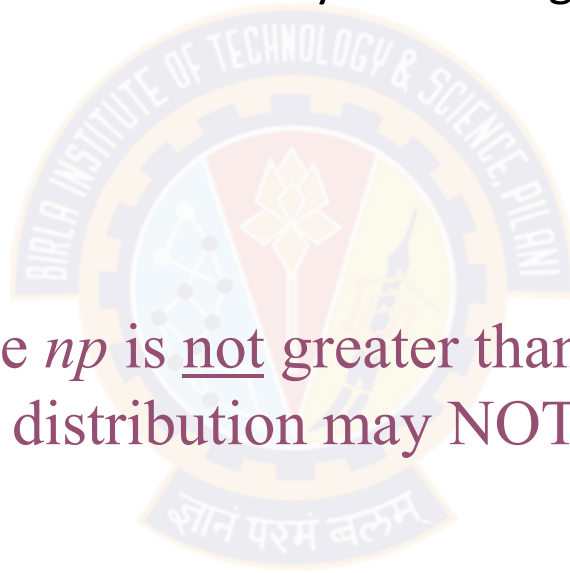
2. Thirty-six percent of people in the United States own a dog. You randomly select 25 people in the United States and ask them if they own a dog.

Solution:

$$np = (25)(0.36) = 9$$

$$nq = (25)(0.64) = 16$$

Because np is not greater than 15, the normal distribution may NOT be used.



Example 18:

Thirty-one percent of the seniors in a certain high school plan to attend college. If 50 students are randomly selected, find the probability that less than 14 students plan to attend college by normal approximation.

Solution:

$$np = (50)(0.31) = 15.5$$

$$nq = (50)(0.69) = 34.5$$

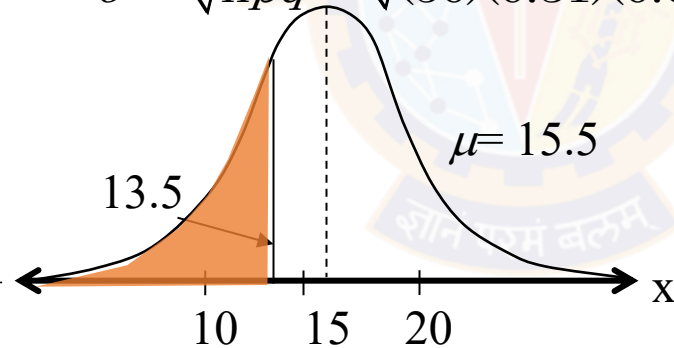
$$P(x < 13.5) = P(z < -0.61) = 0.2709$$

Correction for continuity

$$z = \frac{x - \mu}{\sigma} = \frac{13.5 - 15.5}{3.27} = -0.61$$

The variable x is approximately normally distributed with $\mu = np = 15.5$ and

$$\sigma = \sqrt{npq} = \sqrt{(50)(0.31)(0.69)} = 3.27.$$



The probability that less than 14 plan to attend college is 0.2709.

Example 19:

If 62% of all clouds seeded with silver iodide show spectacular growth, What is the probability that among 40 clouds seeded with silver iodide at most 20 will show spectacular growth by normal approximation?

Soln: Here $n = 40$, $p = 0.62$, $q = 0.38$,
 $\mu = np = 24.8$, $\sigma^2 = npq = 9.424$, $\sigma = 3.0699$
 $np \geq 15$ and $nq \geq 15$

Therefore, we will use normal approximation to binomial distribution.

$$P(x \leq 20.5) = ?$$

$$\text{When } x = 20.5, \quad Z = \frac{x - \mu}{\sigma} = \frac{20.5 - 24.8}{3.0699} = -1.40$$

$$\begin{aligned} \therefore P(x \leq 20.5) &= P(Z \leq -1.40) \\ &= F(-1.40) = 0.808 \end{aligned}$$

Self - study



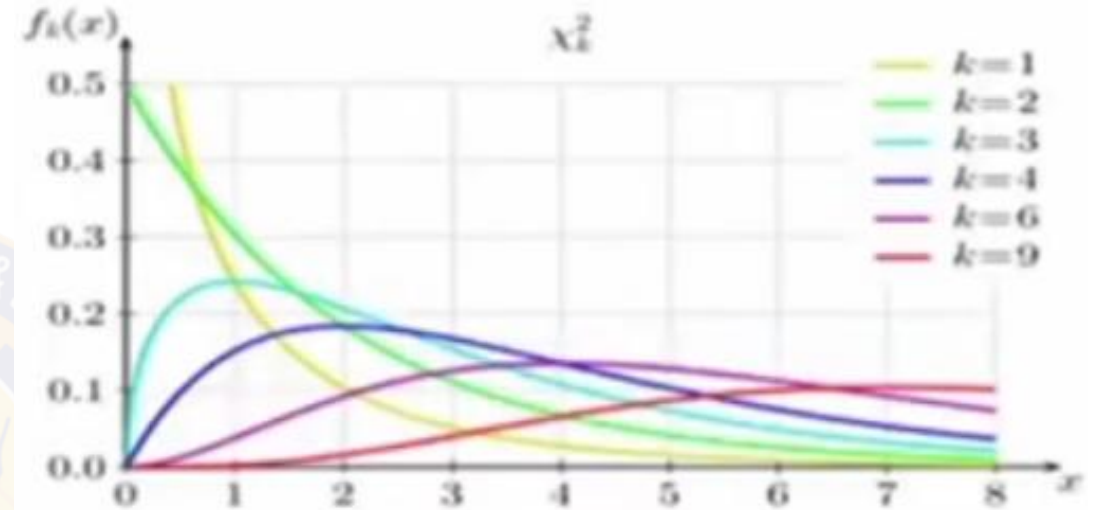
Introduction to Chi- Square distribution

If $z_1, z_2, \dots, z_k$ are $N(0,1)$ and $y = z_1^2 + z_2^2 + \dots + z_k^2$
then y follows χ^2 with k dof.

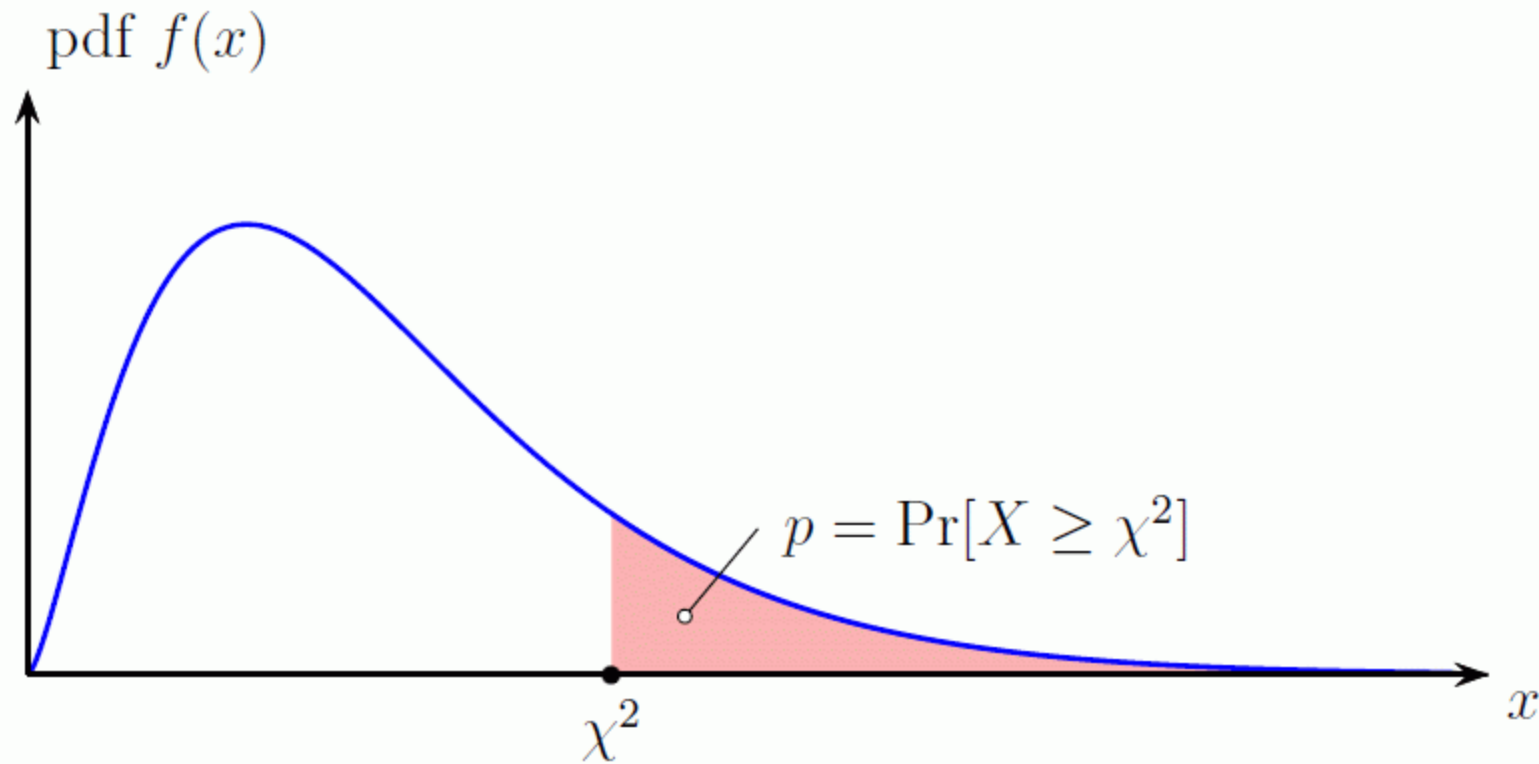
- Chi-Square distribution pdf is given by

$$f(x) = \begin{cases} \frac{1}{2^{k/2} \Gamma(\frac{k}{2})} x^{\frac{k}{2}-1} e^{-\frac{x}{2}} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

- where k is the parameter of distribution, also known as degrees of freedom.



Introduction to Chi- Square distribution



Introduction to Chi- Square distribution

- ❖ Chi-Square distribution curve is not symmetrical, and hence not a normal curve.
- ❖ Chi-Square varies from 0 to ∞ (Curve lies entirely in first quadrant).
- ❖ It depends only on degrees of freedom k .
- ❖ It is very important in estimation and hypothesis testing.
- ❖ It is used in sampling distributions of the sample variance, analysis of variance.
- ❖ It is used as a measure of goodness of fit.

Introduction to t-distribution

Use: If we come across the variable of the form “normal by square root of chi-square upon dof”

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} = \frac{\sqrt{n}(\bar{x} - \mu)}{\sqrt{s^2}}$$

$$t = \frac{\sqrt{(\bar{x} - \mu) / \sqrt{(\sigma^2 / n)}}}{\sqrt{\left(\frac{(n-1)s^2}{\sigma^2} \right) / \frac{1}{(n-1)}}}$$

Where Numerator follows Z distribution and denominator follows chi-Square with n-1 dof.

$$t = \frac{Z}{\sqrt{\chi_{n-1}^2 / n - 1}}$$

$$t_k = \frac{Z}{\sqrt{\chi_k^2 / k}}$$

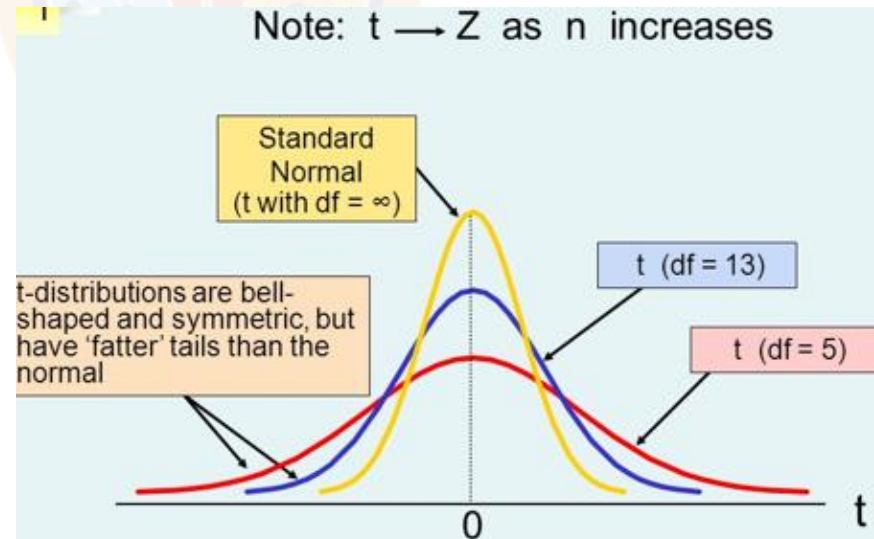
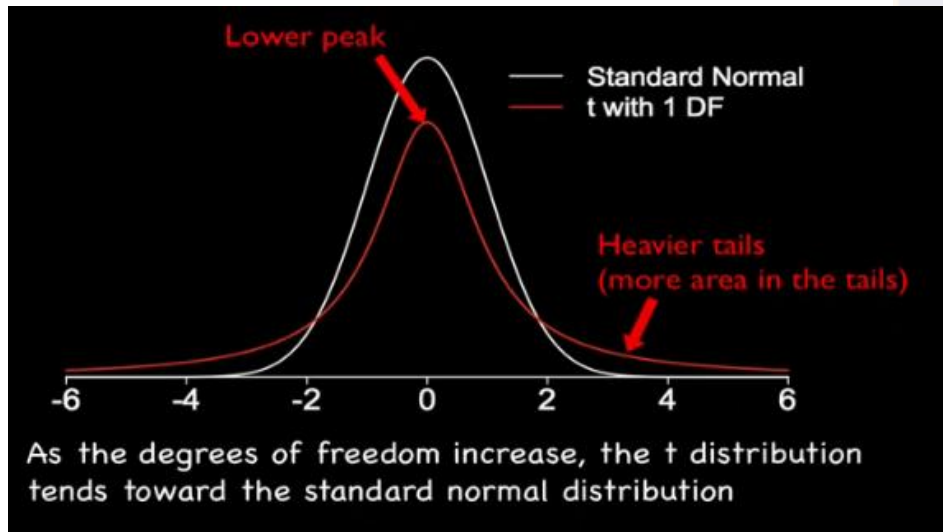
Introduction to t-distribution

If z and χ_k^2 are independent normal and chi-square variables, respectively, then the random variable

$$t_k = \frac{z}{\sqrt{\chi_k^2/k}}$$

follows t distribution with k dof. The pdf of t is

$$f(t) = \frac{\Gamma[\frac{k+1}{2}]}{\sqrt{k\pi}\Gamma(\frac{k}{2})} \frac{1}{[(t^2/k) + 1]^{\frac{k+1}{2}}}$$



Introduction to t-distribution (Sampling distribution of mean when σ unknown)

- The t-distribution, also known as the **Student's t-distribution**, is a type of probability distribution that is similar to the normal distribution with its bell shape but has heavier tails.
- It is used for estimating population parameters for **small sample sizes when**

t - Distribution When σ is unknown and $n < 30$.

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} \text{ with } (n-1) \text{ d.o.f.}$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- Where sample S.D. is calculated by formula

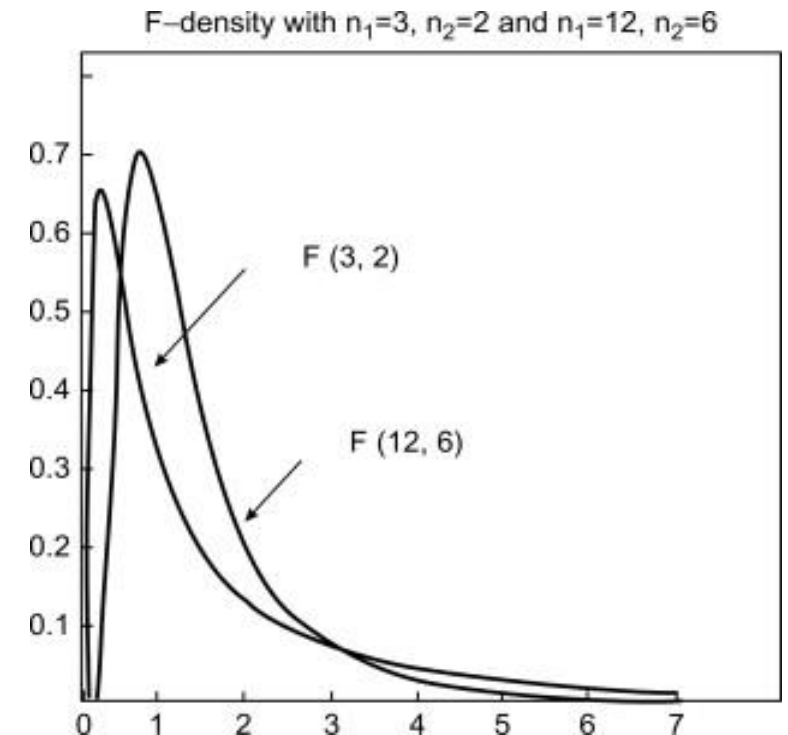
Introduction to F-distribution

Use when come across ratio of two chi- square variables and want to check which population have more variability.

The pdf for a random variable $X \sim F(n_1, n_2)$ is given by

$$f(x) = \frac{\Gamma\left(\frac{n_1 + n_2}{2}\right)}{\Gamma\left(\frac{n_1}{2}\right)\Gamma\left(\frac{n_2}{2}\right)} \left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} x^{\frac{n_1}{2}-1} \left(1 + \frac{n_1}{n_2}x\right)^{-\frac{n_1+n_2}{2}}, x > 0$$

- F- distribution curve lies entirely in first quadrant.
- The F- curve depends not only on the two parameters ϑ_1 and ϑ_2 but also on the order in which they are stated.



Introduction to F-distribution

- ❖ The F-distribution was **developed by Fisher to study the behavior of two variances from random samples taken from two independent normal populations.**
- ❖ In applied problems we may be interested in knowing whether the population variances are equal or not, based on the response of the random samples.
- ❖ If S_1^2 and S_2^2 are variances of independent random sample of size n_1 and n_2 from a normal populations with variances σ_1^2 and σ_2^2 , then

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$

- which follows F –distribution with $\nu_1 = n_1 - 1$ and $\nu_2 = n_2 - 1$ d.o.f.

Practice Problem 1:

In a factory, 8% of all machines break down at least once a year. Use the Poisson approximation to the binomial distribution to determine the probabilities that among 25 machines (randomly chosen in the factory):

- (a) 5 will break down at least once a year;
- (b) at least 4 will break down once a year:
- (c) anywhere from 3 to 8, inclusive, will break down at least once a year.

Solution:

$$\lambda = np = 25 \times 0.08 = 2$$

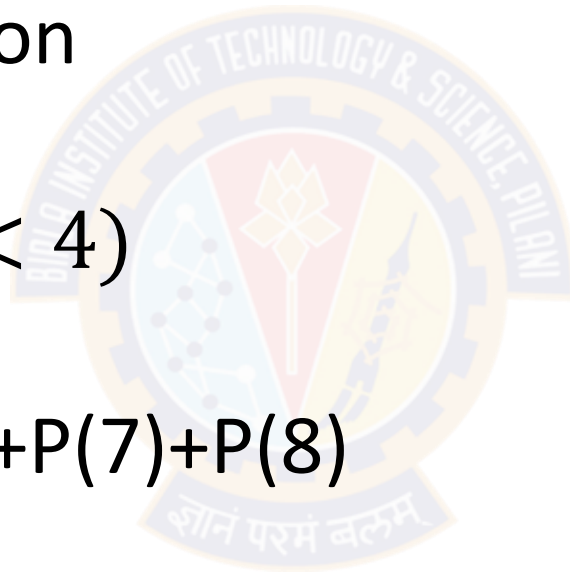
Apply Poisson distribution

a) $P(X=5)$

b) $P(X \geq 4) = 1 - P(X < 4)$

c) $P(3 \leq x \leq$

8) $= P(3) + P(4) + P(5) + P(6) + P(7) + P(8)$



Practice Problem 2:

The average number of trucks arriving on any one day at a truck depot in a certain city is known to be 12. what is the probability that on a given day fewer than nine trucks will arrive at this depot?

Hint : $\lambda = 12$

Apply Poisson distribution

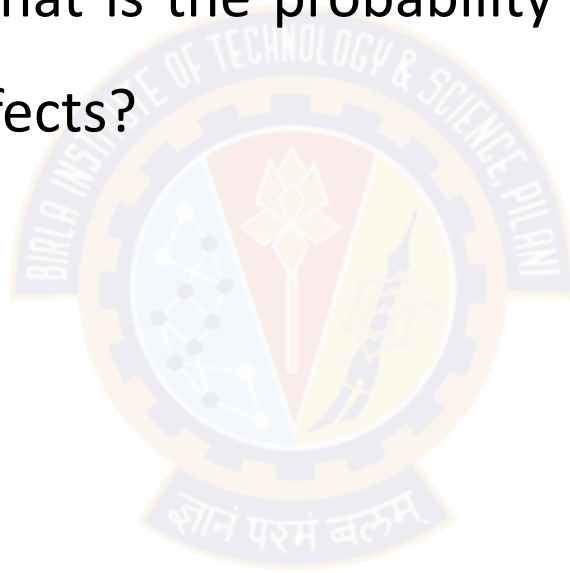
$$P(x < 9) = P(x \leq 8) = P(0) + P(1) + P(2) + P(3) + P(4) + P(5) + P(6) + P(7) + P(8)$$

Practice Problem 3:

A certain kind of sheet metal has on the average, five defects per 10 square feet. If we assume a Poisson distribution, what is the probability that a 15-square-foot sheet of the metal will have at least six defects?

Hint : $\lambda = (15)5/10=7.5$

$$P(X \geq 6) = 1 - P(X < 6)$$



Practice Problem 4:

What is the probability that a standard normal variate Z will be

- (i) greater than 1.09
- (ii) less than -1.65
- (iii) lying between -1.00 and 1.96
- (iv) lying between 1.25 and 2.75

Solution :

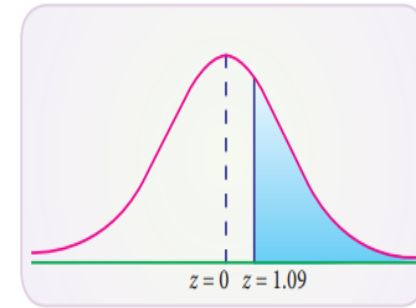
- (i) greater than 1.09

The total area under the curve is equal to 1, so that the total area to the right $Z = 0$ is 0.5 (since the curve is symmetrical).

The area between $Z = 0$ and 1.09 (from tables) is 0.3621

$$P(Z > 1.09) = 0.5000 - 0.3621 = 0.1379$$

The shaded area to the right of $Z = 1.09$ is the probability that Z will be greater than 1.09.



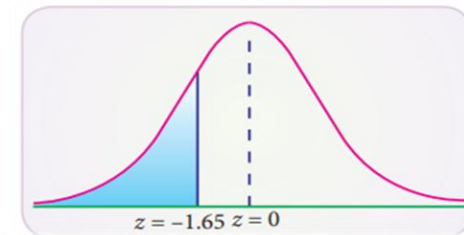
- (ii) less than -1.65

The area between -1.65 and 0 is the same as area between 0 and 1.65.

In the table the area between zero and 1.65 is 0.4505

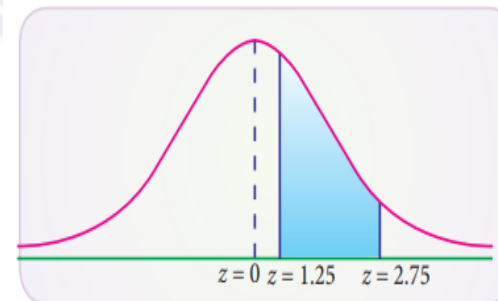
(from the table). Since the area to the left of zero is 0.5,

$$P(Z < -1.65) = 0.5000 - 0.4505 = 0.0495$$



Solution:

- (iii) lying between -1.00 and 1.96
- The probability that the random variable Z is between -1.00 and 1.96 is found by adding the corresponding areas :
- Area between -1.00 and 1.96 = area between $(-1.00$ and $0)$ + area between $(0$ and $1.96)$ $P(-1.00 < Z < 1.96) = P(-1.00 < Z < 0) + P(0 < Z < 1.96) = 0.8163$
- (iv) lying between 1.25 and 2.75 :
- Area between $Z = 1.25$ and 2.75 = area between $(z = 0$ and $z = 2.75)$ – area between $(z=0$ and $z = 1.25)$
- $P(1.25 < Z < 2.75) = P(0 < Z < 2.75) - P(0 < Z < 1.25) = 0.1026$



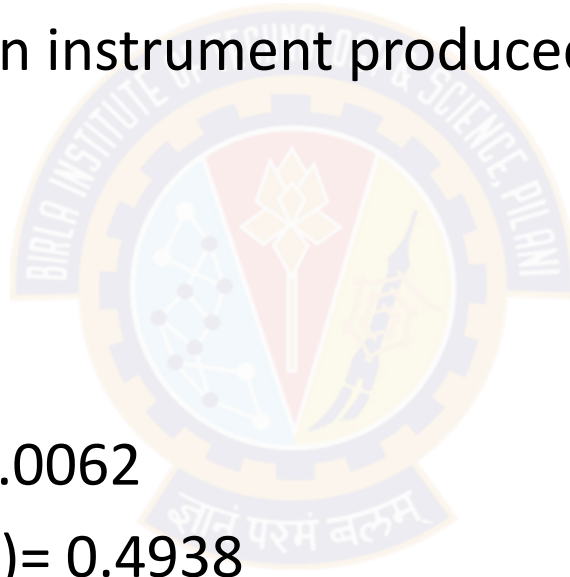
Practice Problem 5:

The length of life of an instrument produced by a machine has a normal distribution with a mean of 12 months and a standard deviation of 2 months. Find the probability that an instrument produced by this machine will last

- a) less than 7 months.
- b) between 7 and 12 months.

Solution: a) $P(x < 7) = P(z < -2.5) = 0.0062$

Hint: b) $P(7 < x < 12) = P(-2.5 < z < 0) = 0.4938$



Practice Problem 6:

A computer crashes on average once every 4 months;

- a) What is the probability that it will not crash in a period of 4 months?
- b) What is the probability that it will crash once in a period of 4 months?
- c) What is the probability that it will crash twice in a period of 4 months?
- d) What is the probability that it will crash three times in a period of 4 months?

Hint:
$$P(x) = \frac{(4)^x e^{-4}}{x!}$$

Practice Problems :

7. If mean of the binomial distribution is 20 and the number of observations is 30. Find the variance and standard deviation.

Hint: $n = 30$, $np = 20$;

8. The probability of a boy hitting a target is 0.2. How Many times must he fire so that the probability of his hitting the target at least once is greater than $2/3$.

Hint: $p = 0.2$; $P(X \geq 1) = 2/3$, find n .

9. A box of candies has many different colors in it. There is a 15% chance of getting a pink candy. What is the probability that exactly 4 candies in a box are pink out of 10?

Hint: $n = 10$, $p = 0.15$, $q = 0.85$, $x = 4$, **Solution:** $P(x=4) = 0.04$;

IMP Note to Self





Thank You!